

CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

Aim:

To write a Prolog program to implement **pattern matching** between two lists.

Objective:

To check whether a given pattern matches a sequence of elements using Prolog's **unification and recursion**.

Algorithm:

1. ☐ Start the program.
2. ☐ Read the pattern and input list.
3. ☐ Compare the first element of the pattern with the first element of the input.
4. ☐ If they match, recursively compare the remaining elements.
5. ☐ If both lists become empty, the pattern matches successfully.
6. ☐ If any element does not match, return false.
7. ☐ Stop.

Prolog Program:

```
% Pattern Matching Program  
pattern_match([], []).  
pattern_match([H1|T1], [H2|T2]) :-  
    H1 = H2,  
    pattern_match(T1, T2)..
```

Output

Result:

Thus, the Prolog program successfully implements **pattern matching using unification and recursion**.